

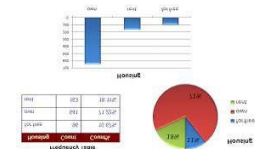
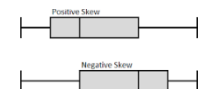
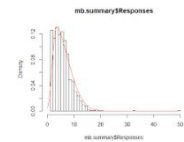
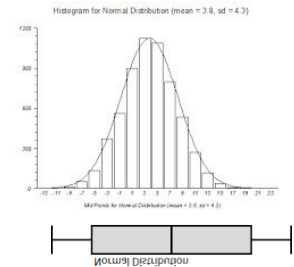
Choosing Your Test

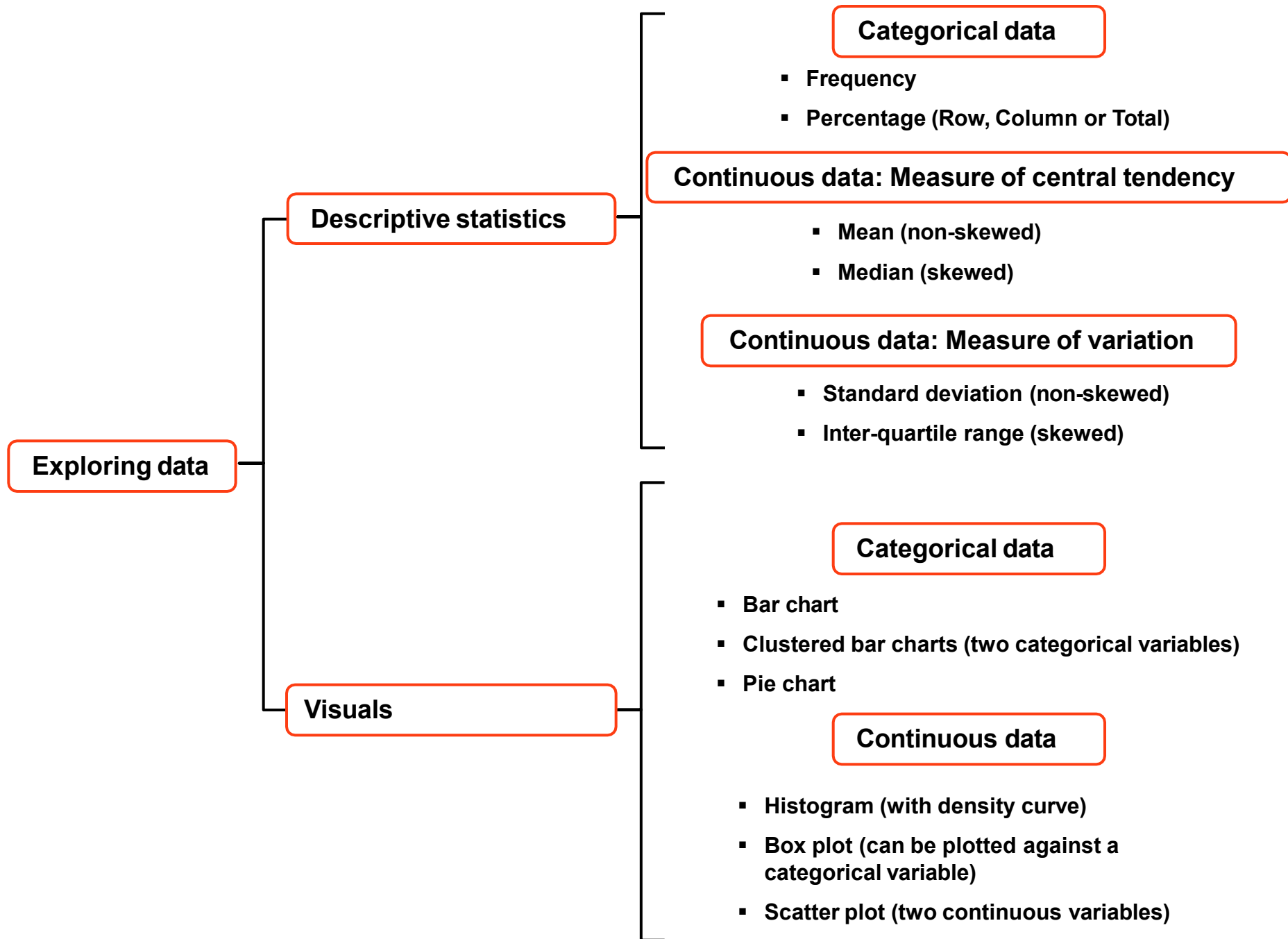
RELATIONSHIP OR DIFFERENCE

A solid orange horizontal bar at the bottom of the slide.

Descriptive statistics and Visuals

- ▶ Dependent on measurement type and shape of distribution
 - ▶ For continuous (Normally distributed) variables
 - ▶ Mean, standard deviation, count
 - ▶ Histograms, dot plots, box plots, scatter plots
 - ▶ For continuous (skewed) variables
 - ▶ Median, inter-quartile range, count
 - ▶ Histograms, dot plots, box plots, scatter plots
 - ▶ For categorical variables
 - ▶ Frequency counts, percentages
 - ▶ One-way tables, two-way tables
 - ▶ Bar charts, pie charts





Deciding Normality

Inspect Visuals

Histogram with
density curve

QQPlot

Calculate standardised skew and Kurtosis

Approaching
normality if
between ± 2

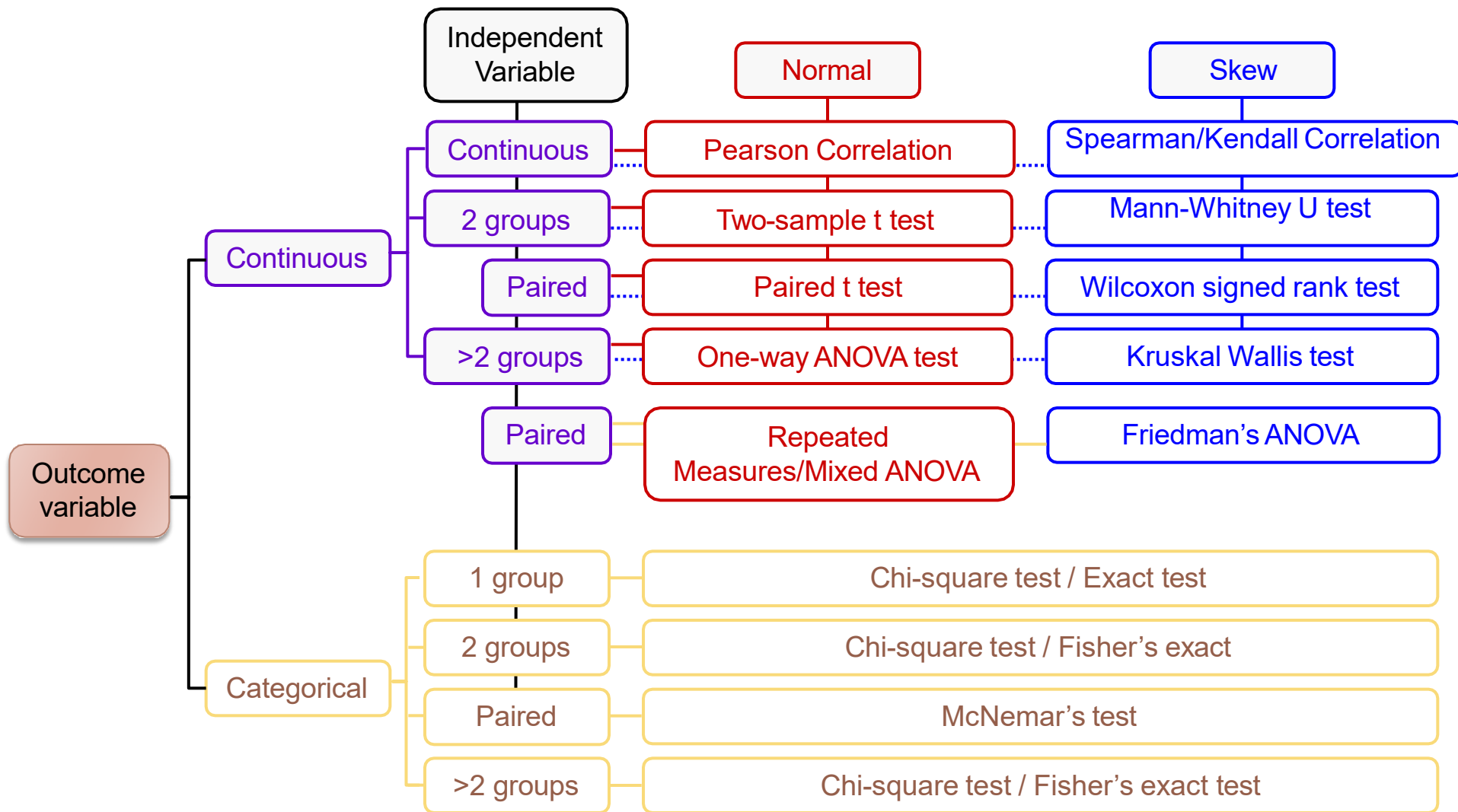
Curran, West
and Finch
(1996)

Create standardised scores for variable (Z scores)

Approaching
normality

- 99 % between ± 3.29 (sample size > 80)
- 95 % ± 1.96 (sample size ≤ 80)
- Field, Miles and Field (2012)

Which test?



Parametric Difference Tests – Pre-Check

T-Test

- Levene' Test
 - Non-significant result
 - variances homogenous in groups
 - (set var.equal=TRUE)
 - Significant result
 - variances heterogeneous
 - (set var.equal=FALSE)
 - Report Welch's Independent Samples t-test

ANOVA

- Bartlett's test
 - Non-significant result
 - Variance is homogeneous in groups (use Tukey post-hoc)
 - Significant result
 - Variance heterogeneous (use Games Howell post-hoc)



Independent variable/ Number of groups	Outcome(Dependent) variable		
	Continuous and Normally distributed (Parametric)	Continuous and skewed / Ordinal (Non-parametric)	Binary (2 categories)
Continuous	Pearson correlation <pre>cor.test method='Pearson'</pre>	Spearman/Kendall correlation <pre>cor.test method='Spearman'/method='Kendall'</pre>	
2 independent groups	T test <pre>t.test var.equal=TRUE/FALSE</pre> Pre-test using <code>car::leveneTet</code>	Mann-Whitney U test <pre>coin::wilcox.test</pre>	Chi-square test / Fisher's Exact <pre>gmodels::Crosstable fisher=TRUE chisq=TRUE</pre>
Paired (related) sample (2 time points)	Paired t test <pre>t.test paired=TRUE/FALSE var.equal=TRUE/FALSE</pre>	Wilcoxon signed rank test <pre>coin::wilcox.test Paired=TRUE</pre>	McNemar's test <pre>mcnemar.test</pre>
>2 independent groups	One-wayANOVA test <pre>stats::oneway.test, var.equal=TRUE/FALSE</pre> Pre-test using <code>stats::bartlett.test</code> Tukey posthoc: <code>rstatix::tukey_hsd</code> Games-Howell posthoc: <code>rstatix::games_howell_test</code>	Kruskal-Wallis test <pre>kruskal.test</pre> Post-hoc: <code>rstatix::dunn_test</code>	Chi-square test / Fisher's Exact Test <pre>gmodels::Crosstable fisher=TRUE chisq=TRUE</pre>
>2 related samples (>2 time points)	Repeated measuresANOVA <pre>Aov</pre>	Friedman's Test <pre>Friedman.test</pre>	